# Weather Conditions and Hourly Bike-Sharing Demand: A Statistical Analysis


## Abstract

This study investigates whether hourly bike rentals are lower during less favorable weather conditions than during clear or partly cloudy conditions. The analysis uses 17,379 hourly observations from the UCI Bike Sharing Dataset. Weather category 1 represents clear or partly cloudy conditions, while categories 2–4 represent less favorable conditions. A Welch's independent-samples t-test was used to compare mean hourly rentals between the two groups. An ordinary least squares (OLS) regression model was also used as a robustness analysis, controlling for hour of the day, season, working-day status, and year, with heteroscedasticity-consistent HC3 standard errors. The mean number of rentals was 204.87 during clear or partly cloudy conditions and 159.99 during less favorable conditions. The estimated difference was −44.88 rentals per hour (95% confidence interval: −50.24 to −39.52; two-sided $p < .001$). The adjusted regression coefficient was −37.89 (95% confidence interval: −41.29 to −34.48; $p < .001$). Both analyses support an association between less favorable weather and lower hourly bike-sharing demand. However, the observational design does not establish causation.


## 1. Introduction

Bike-sharing systems provide a convenient mode of transportation for daily commuting and recreational travel. Understanding the factors associated with bike rental demand can help operators plan bicycle availability and improve service efficiency.

Weather conditions may be associated with people's decisions to use bicycles. Clear or partly cloudy conditions may be more suitable for cycling than conditions involving mist, rain, snow, or other adverse weather.

The present study examines this relationship using hourly records from the UCI Bike Sharing Dataset.

### 1.1 Research question

Among the hourly records in the bike-sharing dataset, is the mean number of bike rentals lower during less favorable weather conditions (weather categories 2–4) than during clear or partly cloudy conditions (category 1)?

### 1.2 Research hypotheses

**Null hypothesis ($H_0$):** The population mean number of hourly bike rentals is equal under less favorable weather conditions and clear or partly cloudy conditions.

**Alternative hypothesis ($H_1$):** The population mean number of hourly bike rentals is lower under less favorable weather conditions than under clear or partly cloudy conditions.

### 1.3 Research objective

The objective is to compare mean hourly bike rentals between the two weather groups and assess whether the observed association remains after accounting for selected temporal and calendar-related variables.

## 2. Methodology

### 2.1 Dataset

The study uses the Bike Sharing Dataset from the UCI Machine Learning Repository. The hourly dataset (`hour.csv`) contains 17,379 observations.

Each observation represents an hourly record and includes information about bike rentals, weather, time, season, and working-day status.

**Dataset source:** UCI Machine Learning Repository, Bike Sharing Dataset.

https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset

### 2.2 Population and sample

The study population consists of hourly observations represented by the bike-sharing dataset. The final eligible analytical sample contained 17,379 observations.

The sample was divided into two groups according to the recorded weather category:

- **Group A — Clear or partly cloudy:** Weather category 1, with 11,413 observations.
- **Group B — Less favorable weather:** Weather categories 2–4, with 5,966 observations.

The analysis used the recorded hourly rental count (`cnt`) as the outcome variable.

## 2.3 Variables

| Variable | Role | Description |
|---|---|---|
| `cnt` | Outcome | Total number of bike rentals per hour |
| Weather group | Main explanatory variable | Clear/partly cloudy versus less favorable weather |
| `hr` | Control | Hour of the day |
| `season` | Control | Recorded season |
| `workingday` | Control | Whether the day was classified as a working day |
| `yr` | Control | Year indicator |

## 2.4 Statistical analysis

The primary analysis compared the mean hourly rental counts between the two weather groups using Welch's independent-samples t-test. This test does not require the two groups to have equal population variances.

The mean difference was calculated as:

**Mean difference = Group B mean − Group A mean**

A 95% confidence interval was calculated for the difference. Cohen's d was also reported as a standardized measure of the difference between the groups.

The supplied analysis reports a two-sided p-value for the Welch test. Statistical significance was evaluated at the 5% level.

For robustness, an OLS regression model was fitted with hourly rental count as the dependent variable. The model included the less-favorable-weather indicator and categorical controls for hour, season, working-day status, and year. HC3 heteroscedasticity-consistent standard errors were used.

The robustness model assessed whether the weather association remained after adjustment for these variables. It was not interpreted as proof of a causal effect.

# 3. Results

## 3.1 Descriptive statistics

The final analytical sample consisted of 17,379 observations.

| Weather group | Sample size | Mean rentals per hour | Standard deviation |
|---|---|---|---|
| Clear/partly cloudy | 11,413 | 204.87 | 189.49 |
| Less favorable weather | 5,966 | 159.99 | 160.74 |

The average hourly rental count was lower in the less favorable weather group.

The difference between the group means was approximately 44.88 fewer rentals per hour in the less favorable weather group.

## 3.2 Primary hypothesis test

Welch's independent-samples t-test produced the following results:

| Statistic | Result |
|---|---|
| Mean difference (Group B − Group A) | −44.88 |
| 95% confidence interval | [−50.24, −39.52] |
| Welch's t-statistic | −16.413 |
| Degrees of freedom | 13,935.52 |
| Two-sided p-value | $5.66 \times 10^{-60}$ |
| Cohen's d | −0.249 |

The p-value was substantially below the 0.05 significance threshold. The 95% confidence interval for the mean difference was entirely below zero.

Therefore, the results provide strong statistical evidence of a difference in mean hourly rentals between the two weather groups, with fewer rentals observed during less favorable weather conditions.

The absolute value of Cohen's d was approximately 0.25, indicating a small standardized difference between the groups.

The estimated difference of 44.88 rentals represents a difference between group averages, not a prediction that every hour with less favorable weather will have exactly 44.88 fewer rentals.

## 3.3 Robustness analysis

The OLS regression included controls for hour of the day, season, working-day status, and year.

| Statistic | Result |
|---|---|
| Adjusted weather coefficient | −37.89 |
| 95% confidence interval | [−41.29, −34.48] |
| Reported p-value | $2.57 \times 10^{-105}$ |
| Number of observations | 17,379 |
| R-squared | 0.647 |
| Adjusted R-squared | 0.646 |

The estimated coefficient for less favorable weather was −37.89. Holding the included control variables constant, the model estimated approximately 37.89 fewer rentals per hour for the less favorable weather group.

The confidence interval remained entirely below zero, and the reported p-value was below 0.001. The negative association therefore remained after adjustment for the included variables.

The R-squared value of 0.647 indicates that the fitted model explained approximately 64.7% of the variation in hourly rental counts within the analyzed sample. This describes the model's fit; it does not mean that the included variables causally explain 64.7% of rental demand.

## 3.4 Comparison of the primary and robustness analyses

Both analyses produced negative estimates:

- The unadjusted difference was −44.88 rentals per hour.
- The adjusted regression coefficient was −37.89 rentals per hour.

The adjusted estimate was smaller in magnitude than the unadjusted difference, but its confidence interval remained below zero.

The consistency in direction supports the conclusion that the observed relationship was not eliminated by adjustment for the measured hour, season, working-day status, and year variables.

## 4. Discussion

The findings support the study's main expectation that hourly bike rentals are lower during less favorable weather conditions than during clear or partly cloudy conditions.

The primary analysis found a mean difference of approximately 44.88 rentals per hour. The robustness model continued to show a negative association after accounting for selected temporal and calendar-related variables.

One possible interpretation is that less favorable weather may make cycling less comfortable or convenient, potentially influencing people's willingness to rent bicycles. However, this study did not directly measure individual motivations, so this explanation should be treated as a possible interpretation rather than a demonstrated mechanism.

The adjusted coefficient was smaller than the unadjusted mean difference. This suggests that differences associated with the included control variables account for some of the initial difference in group averages. Nevertheless, the negative adjusted association remained statistically strong.

These findings may be useful for bike-sharing operators when considering weather-related demand patterns. However, operational decisions should also consider actual weather forecasts, local conditions, availability constraints, and other demand factors.

## 5. Limitations

Several limitations should be considered when interpreting the results.

**1. Observational design:** The study uses existing observational data. It establishes an association between weather categories and rental counts but does not establish that weather alone caused the difference.

**2. Potential confounding:** The regression adjusts for hour, season, working-day status, and year, but other relevant factors may remain unmeasured. Examples include holidays, special events, local travel patterns, and detailed weather characteristics.

**3. Temporal dependence:** Hourly observations collected over time may not be statistically independent. The robustness output reports a Durbin–Watson statistic of 0.448, suggesting substantial positive residual autocorrelation. HC3 standard errors address heteroscedasticity but do not, by themselves, correct for serial correlation. Consequently, the reported regression confidence interval and p-value may overstate the precision of the adjusted estimate. A model that explicitly accounts for temporal dependence would be a useful additional sensitivity analysis.

**4. Grouped weather categories:** Combining weather categories 2–4 into a single group simplifies comparison but hides possible differences among misty, rainy, snowy, or other adverse conditions.

**5. Generalizability:** The results describe the observations in this dataset. They may not apply equally to every bike-sharing system, city, season, or population.

**6. Rental counts are variable:** Hourly counts have substantial variability within both weather groups, as indicated by their standard deviations. Weather group membership alone does not determine demand in a particular hour.

## 6. Preregistration and Deviations

The analysis was conducted using the stated weather-group comparison and the supplied primary and robustness analysis outputs.

| Item | Status |
|---|---|
| Research question and weather-group comparison | Defined in the preregistration |
| Primary statistical analysis | Welch's t-test; the supplied output reports a two-sided p-value |
| Robustness analysis | OLS regression with specified calendar and time controls and HC3 standard errors |
| Changes to the preregistered plan | Must be confirmed against the final analysis implementation |

No specific change to the preregistered plan has been identified from the supplied results alone. Before submission, the Deviations table should explicitly state that no deviations occurred only if this has been verified. Any difference between the preregistered plan and the analysis actually performed must be documented, including the reason for the change.

The two-sided p-value reported by the primary test should also be checked against the preregistered testing procedure. If a one-sided test was preregistered but a two-sided test was performed, that difference should be disclosed rather than silently changed.

## 7. Conclusion

This study examined whether mean hourly bike rentals were lower during less favorable weather conditions than during clear or partly cloudy conditions.

Across 17,379 hourly observations, the mean rental count was 204.87 under clear or partly cloudy conditions and 159.99 under less favorable conditions. The estimated difference was −44.88 rentals per hour, with a 95% confidence interval from −50.24 to −39.52. The Welch test produced a two-sided p-value of approximately $5.66 \times 10^{-60}$.

The robustness regression also produced a negative weather coefficient of −37.89 rentals per hour after adjustment for hour, season, working-day status, and year.

Overall, both analyses support the conclusion that less favorable weather conditions were associated with lower hourly bike rentals in this dataset. The results are statistically strong, but their causal interpretation is limited by the observational design and the possibility of temporal dependence in the data. Future work could use methods that account explicitly for time-series dependence and examine individual weather categories separately.

## 8. Reference

UCI Machine Learning Repository. *Bike Sharing Dataset*.
https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset